

ECON 351 · Midterm Simulation

Practice Test 1

24 Questions | 24 points (1 pt each)

FORMAT (matches your teacher's breakdown):

PART A · Chapter 4 (12 questions)

- 2 Consumer Problems
- 2 Labor-Leisure
- 2 Intertemporal Consumption
- 3 Elasticity / Types of Goods
- 1 Budget Line
- 2 Assumptions / Preferences

PART B · Chapter 9 (12 questions)

- 5 Insurance Problems
- 4 Risk Attitudes
- 2 Certainty Equivalent & Risk Premium
- 1 Diversification

Problems from HW, sample questions, and mock midterm (screenshots).

Use Formula_Sheet.pdf for reference.

PART A

CHAPTER 4

12 questions

Exercise 3 - Optimal Consumption

3.1 Manuel has the following utility $U(q_1, q_2) = 3q_1^{0.6}q_2^{0.4}$. What values of q_1 and q_2 maximize his utility?

3.2 Now suppose prices are $p_1 = 5$ and $p_2 = 10$ and that Manuel has \$100. Write Manuel's budget constraint and graph it with q_1 in the x-axis. What is the slope? And the intercept? What is the maximum amount of q_1 he can purchase? And the maximum amount of q_2 he can purchase?

3.3 Next, we will find out the best use Manuel could make of his money, given his preferences. Based on Manuel's preferences, let's figure out the ratio between q_1 and q_2 . What is the ratio of the two marginal utilities (MRS)? (remember that MU_1 is in the numerator)

3.4 Now we have the ratio of how Manuel trades both goods. We will equalize it to the ratio of both prices. That is, how Manuel trades both goods needs to equal how the market trades them. In the optimal bundle, how do the marginal utilities relate to market prices?

3.5 Finally, use the previous relationship to substitute in the budget constraint and find the optimal quantities of q_1 and q_2 , which we denote by q_1^* and q_2^* .

Question 26:

There are two goods, apples and bananas. The price of apples is $P_A = \$2$, and the price of bananas is $P_B = \$3$. A consumer has \$120 to spend, and his utility function is $U(A, B) = 2A^2B^3$.

a) Write the expression that represents the consumer's budget constraint. Graph the budget constraint, and label it "BC" (apples should be on the horizontal axis). What is the slope of the budget constraint?

b) Find the consumption basket that maximizes the consumer's utility. Plot this point on the graph, and label it as "Point X". Draw the indifference curve that goes through Point X, and label it "IC" (the indifference curve can be an approximation, it does not have to be exact). What is the slope of the indifference curve at Point X?

Question 27:

Adam derives utility from consuming apples and bananas. His marginal utility from consuming apples is

$$MU_A = 5A^4B^3,$$

while his marginal utility from consuming bananas is

$$MU_B = 3A^5B^2,$$

where A is the quantity of apples and B is the quantity of bananas. We know that the price of apples is $P_A = 6$, the price of bananas is $P_B = 2$ and Adam's income is $I=240$. Given this information, when Adam optimally chooses his consumption, he buys how many units of apples?

(A) A=65

(B) A=45

(C) A=25

(D) A=35

(E) A=55

Question 28:

Monica only consumes goods A and B. Suppose that her marginal utility from consuming good A is equal to $1/Q_A$, and her marginal utility from consuming good B is $1/Q_B$. If the price of A is \$0.50, the price of B is \$4.00, and the Monica's income is \$120.00, how much of good A will she purchase?

(A) 0

(B) 12

(C) 24

(D) 48

(E) 120

[Q021]

Maria: 16 hrs/day, $w=\$20$, $V=\$80$, $U=c^{0.5} l^{0.5}$. Find I^* , L^* , c^* .
(Course-style)

[Q022]

Same setup, $w=\$30$. Find I^* , L^* , c^* . More or less work?
(Course-style)

Question 38:

A consulting company researched the market for apples in California, and concluded the following: apples and bananas are complements, and apples are an inferior good.

Define the following:

Q_A : quantity of apples demanded

P_A : price of Apples

P_B : price of bananas

I : income

From the options below, mark the only demand function that can possibly represent the demand for apples in California.

(A) $Q_A = 100 - 3P_A + 3I - 5P_B$

(B) $Q_A = 200 - P_A + 2I + 4P_B$

(C) $Q_A = 110 - 2P_A - 4I + 2P_B$

(D) $Q_A = 250 - 4P_A - 5I - 3P_B$

(E) $Q_A = 150 + P_A + I + P_B$

Question 39:

The demand for apples (Q_A) is given by the function

$$Q_A = 784 - 2P_A - 3I - 4P_O$$

where P_A is the price of apples, I is the income level, and P_O is the price of oranges.

Suppose $P_A = 2$, $I = 100$, and $P_O = 20$.

a) Compute the price elasticity of demand for apples.

b) Compute the income elasticity of demand for apples. Are apples a normal good or inferior good?

c) Compute the cross-price elasticity of demand for apples. Are apples and oranges substitutes or complements?

Question 40:

The income elasticity of demand for good X is -2. Due to a financial crisis, the income of consumers declined by 2%. How did the financial crisis affect the quantity demanded for X?

(A) The quantity demanded increased 2%

(B) The quantity demanded decreased 2%

(C) The quantity demanded increased 4%

(D) The quantity demanded decreased 4%

(E) The quantity demanded did not change

Question 41:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 50%, that is, $r = 0.5$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ? Is this consumer saving or borrowing? How much?

Question 42:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 5%, that is, $r = 0.05$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ?
Is this consumer saving or borrowing? How much?

Question 24:**[Show your work]**

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 60$ in the first period and $I_2 = 125$ in the second period. The consumer can borrow or save money at the interest rate of 25%, that is, $r=0.25$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

The consumer's inter-temporal budget constraint is

- (A) $q_1 + 0.8q_2 = 160$
- (B) $q_1 + q_2 = 185$
- (C) $q_1 + 1.25q_2 = 216.25$
- (D) $q_1 + 0.75q_2 = 160$
- (E) $q_1 + 0.25q_2 = 200$

Question 25:**[Show your work]**

Consider all the information from Question 24. In the first period, the consumer will optimally purchase

- (A) $q_1 = 0$
- (B) $q_1 = 100$
- (C) $q_1 = 60$
- (D) $q_1 = 80$
- (E) $q_1 = 120$

Exercise 4 - Price Elasticity

Here, we will review the concept of elasticity and go through some examples. Price elasticity of demand is how much quantity demanded changes when the price goes up by 1%. Reminder $E_p = \frac{P}{Q} \frac{\partial Q}{\partial P}$

Take the following demand function $q = 100 - 5p$.

4.1 Calculate the quantity demanded when $p = 5$ and when $p = 10$. What happens to the quantity demanded when the price increases? Given your answer, do you expect the sign of price elasticity to be positive or negative?

4.2 Calculate the price elasticity when $p = 5$. Interpret the elasticity.

4.3 Calculate the quantity demanded when $p = 5.05$.

4.4 Note that $p = 5.05$ corresponds to a 1% price increase with respect to $p = 5$. Compare this % increase with the % change in the quantity demanded. That is, calculate $\Delta Q / Q(p = 5) * 100$. How does this compare to the price elasticity you calculated before?

4.5 Redo points 4.2-4.4 but now calculate the elasticity at $p = 10$ and compare the % change in quantity demanded when $p = 10.1$. How does the % change in quantity compared to the elasticity at $p = 10$?

4.6 EXTRA In the previous questions, you have calculated the elasticity using the formula and also by "brute force". In these cases, the values coincide, but in other cases, there would be a small gap between both methods. Why is that the case, and why are they the same here? (Hard question. Tip: it has to do with the basic concept of a derivative and the curvature of the demand function.)

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From the options below, mark the only demand function that can possibly represent the demand for apples in California.

(A) $Q_A = 100 - 3P_A + 3I - 5P_B$

(B) $Q_A = 200 - P_A + 2I + 4P_B$

(C) $Q_A = 110 - 2P_A - 4I + 2P_B$

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Suppose $P_A = 2$, $I=100$, and $P_O = 20$.

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What is the optimal choice of q_1 and q_2 ? Is this consumer saving or borrowing? How much?

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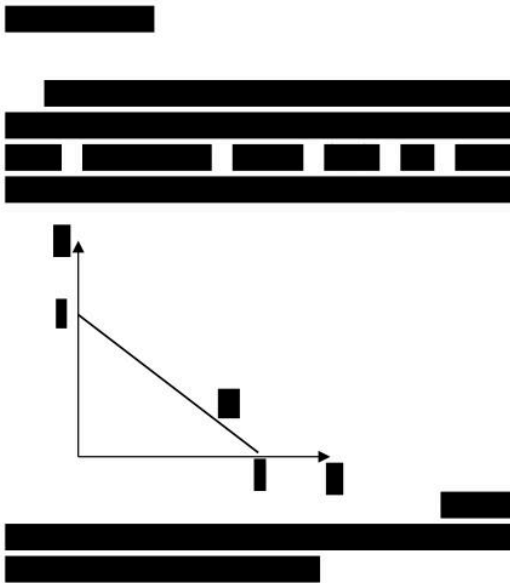
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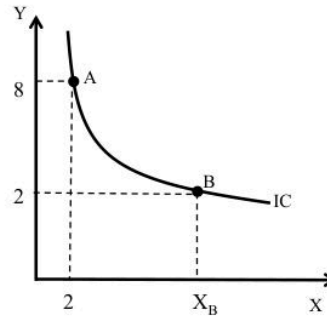
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_____e

Question 9:

Angelique's preferences over goods X and Y can be represented by the following utility function: $u(X, Y) = X^2Y^3$. The graph below represents two baskets, A and B, on the same indifference curve of this consumer.



What do we know about X_B ?

- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

Question 10:

Assume that apples are measured on the horizontal axis and oranges on the vertical axis. If the price of apples increases from \$3 to \$6, and the price of oranges increases from \$2 to \$4, then the budget line will:

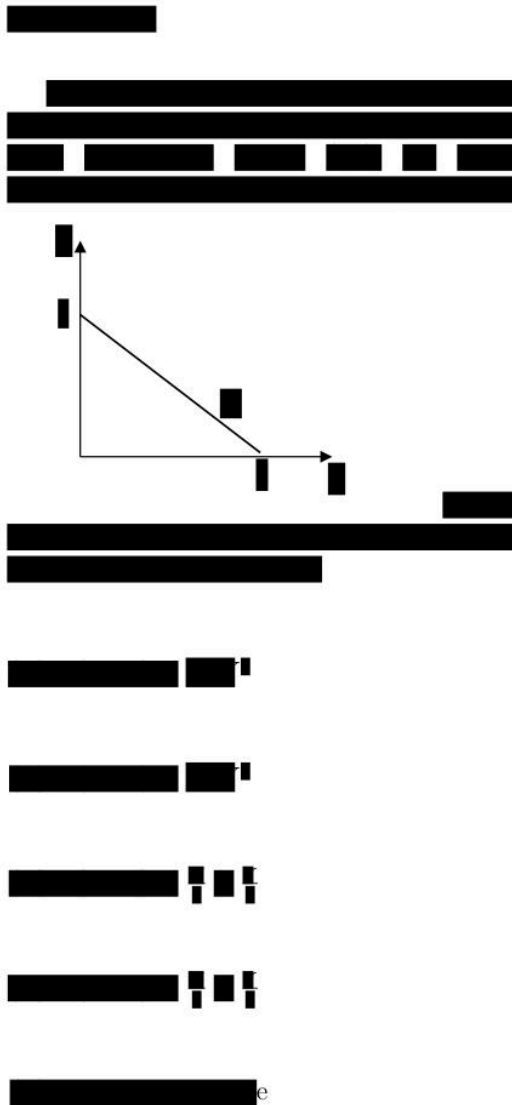
- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).
- (E) become steeper or flatter depending on the relationship between prices and income.

Exercise 2 - Preferences' Assumptions

2.1 Jackie was at the movie theater last night. The two movies available were “Knives out” and “Top Gun Maverick.” She didn’t care which one to see. Does this imply a violation of any assumption on preferences? If so, which one?

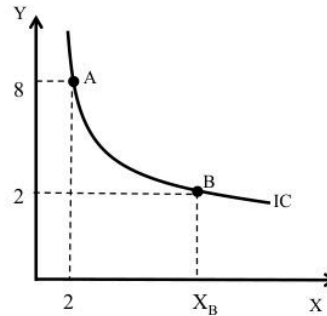
2.2 Again, Jackie goes to the movies. The two movies available are “Avatar” and “The Avengers,” and she says she cannot decide which one to see. Does this imply a violation of any assumption on preferences? If so, which one?

2.3 On a different day, there were three movies available: “Parasite,” “Indiana Jones,” and “Jurassic Park.” Jackie said she rather watch Parasite over Indiana Jones and Indiana Jones over Jurassic Park. But she would also prefer watching Jurassic Park over Parasite. Does this imply a violation of any assumption on preferences (beyond displaying poor taste)? If so, which one?



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Angelique's preferences over goods X and Y can be represented by the following utility function: $u(X, Y) = X^2Y^3$. The graph below represents two baskets, A and B, on the same indifference curve of this consumer.



What do we know about X_B ?

- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

Question 10:

Assume that apples are measured on the horizontal axis and oranges on the vertical axis. If the price of apples increases from \$3 to \$6, and the price of oranges increases from \$2 to \$4, then the budget line will:

- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).
- (E) become steeper or flatter depending on the relationship between prices and income.

PART B

CHAPTER 9

12 questions

Exercise 3 - Insurance

Tom has the following utility from income:

$$u(I) = \sqrt{x}$$

He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house. In this case, the house value becomes \$672,400. AAA offers to fully insure the house for a price P .

- a) What is the maximum insurance premium P that Tom is willing to pay?
- b) How much is the expected loss? (also known as the fair insurance premium)
- c) How much is the risk premium? (difference between the max.premium the consumer is willing to pay and the expected loss)

- c) Compute the certainty equivalent of this investment.
- d) Compute the risk premium associated with this investment.

Question 13:

Cuban is an investor who used all his income to buy a startup company called Shark Aquarium. With probability 50% the value of the company will increase to \$144. With probability 50% the value of the company will go down to \$64. Cuban's utility from income is

$$u(I) = \sqrt{I}.$$

He has no other source of income. Compute the minimum price P such that Cuban would be willing to sell all his shares of Shark Aquarium.

- (A) $P = 94$
- (B) $P = 130$
- (C) $P = 121$
- (D) $P = 104$
- (E) $P = 100$

Question 15:

Tom has the following utility from income: $u(I) = \sqrt{I}$. He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house (in this case, the house value becomes \$672,400). AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Tom is willing to pay?

- (A) 23,010
- (B) 45,375
- (C) 74,000
- (D) 32,375
- (E) 15,600

Question 16:

Using the information from the previous question, how much is the fair insurance premium?

- (A) 174,000
- (B) 72,000
- (C) 130,500
- (D) 43,500
- (E) 168,100

Question 19:

Joe's utility from income is given by

$$u(I) = 8\sqrt{I}.$$

He has a house in LA County. With probability $1/4$, the rain will cause a mudslide and damage his house, decreasing its value to \$250,000. With probability $3/4$, no mudslide occurs and the house retains its original \$1,000,000 value. AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Joe is

Question 18:

Sherman has the following utility from wealth:

$$u(x) = \sqrt{x}.$$

Wealth is the sum of cash and the value of his car. He currently has a Ferrari worth \$230,900 and \$50,000 in cash (risk free). With probability 25%, a paparazzi will cause \$158,400 in damages to the car (in this case, the car value becomes \$72,500). With probability 75% nothing happens, and the car keeps its original value. AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that Sherman is willing to pay? What is the associated risk premium?

insure the car for a price $P=124$. That is, Marcelo pays insurance premium $P=124$, and in case of a crash, Progressive will only pay for half of the repair cost (in case of a crash, Progressive will only pay \$3,174 to Marcelo). If Marcelo buys this partial insurance plan, his expected utility will be:

(A) 94

(B) 98

(C) 96

(D) 95

(E) 97

Question 27:

[Show your work] A consumer has \$1,000 in cash. With probability 90% nothing happens (he does not get sick) and he is able to keep all his money. With probability 10% he gets sick and must go to the doctor. The cost of going to the doctor is \$500.

An insurance company offers the following health insurance plan. The consumer must pay upfront a premium $p = \$100$. If the consumer gets sick, the insurance company pays the doctor, but the consumer must pay a \$275 copay.

The consumer's utility from income is given by $u(I) = \sqrt{I}$. Compute the expected utility of the consumer if he purchases this insurance.

- (A) 29.5
- (B) 30.7
- (C) 28.6
- (D) 30
- (E) 29

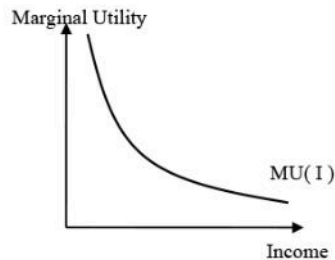
Question 25:

[Show your work] A consumer has the following utility from income: $u(I) = \sqrt{I}$. He bought a car for \$10,000. With probability 75%, a crash will cause \$3,600 in damages to the car (in this case, the car's value becomes \$6,400). AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that the consumer is willing to pay?

- (A) 2,725
- (B) 2,850
- (C) 2,750
- (D) 2,825
- (E) 2,775

Exercise 2 - Attitudes towards risk and fair bets

The following graph represents Mike's **marginal utility** of income (not the utility but the marginal utility):



We know that Mike's income I is greater than \$20, but we do not know his exact income. Alejandro offers Mike the following bet: Alejandro will roll a 6 sided dice (all numbers equally likely). If it is 1 or 2, Mike pays Alejandro \$20. Otherwise, if it is 3, 4, 5 or 6, Alejandro pays Mike \$10.

a) Is this a fair bet?

b) Will Mike accept or reject the bet? Does your answer depend on Mike's initial wealth? (assuming he can always pay the \$20 if he loses)

- a) Compute the expected value of each investment option.
- b) Compute the expected utility of each investment option. Find the best (and worse) investment option for Jim.

Question 3:

Pedro Carroll has the following utility from income: $u(I) = \sqrt{I}$. Currently, he has an income $I = \$169$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Seahawks to win the championship game. If he bets all his money and the Seahawks win, then he ends up with an income $I = \$400$. If he bets all his money and the Seahawks lose, then he ends up with nothing, $I = \$0$. If he does not bet, he keeps his \$169 for sure.

The Los Angeles Seahawks will win the championship game with probability P , and will lose the game with probability $1-P$.

Compute the minimum winning probability P such that Pedro would be willing to bet all his money on the Seahawks.

- (A) $P = 50\%$
- (B) $P = 65\%$
- (C) $P = 70\%$
- (D) $P = 55\%$
- (E) $P = 60\%$

Question 4:

Cam has the following utility from income: $u(I) = \sqrt{I}$. Currently he has an income $I = \$36$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Panthers to win the championship game. If he bets all his money and

the Panthers win, then he ends up with an income I_{Win} . If he bets all his money and the Panthers lose, then he ends up with nothing, $I_{Lose} = \$0$. If he does not bet, he keeps his \$36 for sure.

The Los Angeles Panthers will win the championship game with probability 75%, and will lose the game with probability 25%.

Compute the minimum winning prize I_{Win} such that Cam would be willing to bet all his money on the Panthers.

- (A) $I_{Win} = \$36$
- (B) $I_{Win} = \$48$
- (C) $I_{Win} = \$56$
- (D) $I_{Win} = \$64$
- (E) $I_{Win} = \$72$

**Question 6:**

Anna's utility of income function is $U(I) = I$, while Elsa's utility of income function is $U(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

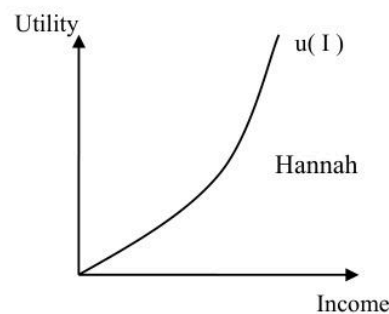
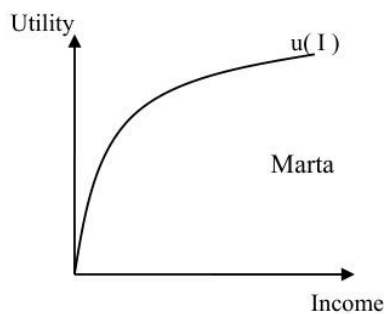
Question 7:

Anna's marginal utility of income function is $MU(I) = I$, while Elsa's marginal utility of income function is $MU(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 8:

The following graphs represent the utility function of consumers Marta and Hannah:

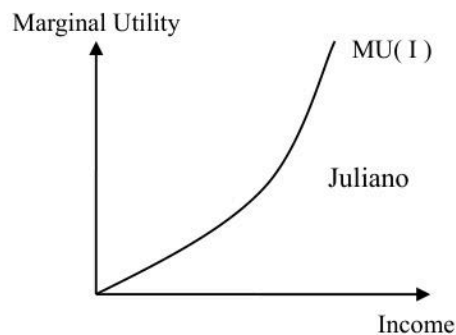
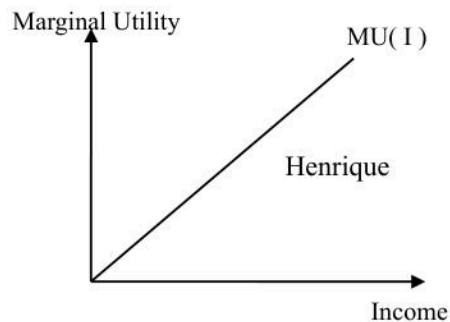


Using the information from the graphs,

- (A) Both Marta and Hannah are risk loving
- (B) Both Marta and Hannah are risk averse
- (C) Marta is risk averse; Hannah is risk loving
- (D) Marta is risk loving; Hannah is risk averse
- (E) Not enough information from the graphs

Question 9:

The following graphs represent the marginal utility of income functions of consumers Henrique and Juliano.



Using the information from the graphs, Henrique is _____ and Juliano is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 10:

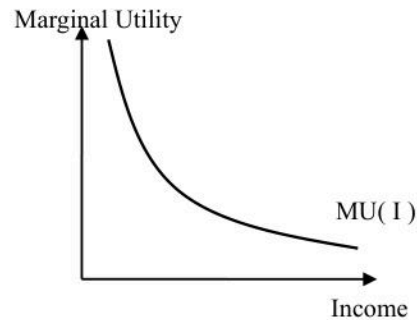
Mike has a constant marginal utility of income. We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough

- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 11:

The following graph represents Mike's marginal utility of income:



We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough
- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 28:

Firmino is a big fan of a soccer team called The Reds. Everyone knows that there is a 75% chance that the The Reds will win the next tournament, and a 25% chance that they will lose. Firmino offers the following bet to each one of his friends: if The Reds win, then the friend must pay \$1 to Firmino. If The Reds lose, then Firmino will pay \$3 to the friend.

The friends Mohamed and Sadio (who are very rich) are each considering whether to accept the bet. Mohamed's utility of income function is

$$U(I) = I^{0.7}$$

while Sadio's utility of income function is

$$U(I) = I^{2.3}$$

Therefore,

- (A) Both Mohamed and Sadio will reject the bet
- (B) Both Mohamed and Sadio will accept the bet
- (C) Mohamed will accept and Sadio will reject the bet
- (D) Mohamed will reject and Sadio will accept the bet
- (E) Both Mohamed and Sadio are indifferent between accepting and rejecting the bet

Question 29:

Assets A and B have positively correlated returns, while assets C and D have negatively correlated returns. John is an investor who has chosen to invest half of his money on asset A and half on asset B. Marc is an investor who has chosen to invest half of his money on asset C and half on asset D. Therefore, which

of the following options is most likely to be true when we compare John and Marc?

- (A) John's investment has a lower variance and John is more risk averse than Marc
- (B) John's investment has a higher variance and John is more risk averse than Marc
- (C) John's investment has a lower variance and John is less risk averse than Marc
- (D) John's investment has a higher variance and John is less risk averse than Marc
- (E) Both investment portfolios have the same variance and the investors are risk neutral.

Question 30:

Mirabel's marginal utility of income function is $MU(I) = 9$, while Bruno's marginal utility of income function is $MU(I) = \frac{3}{I}$. Therefore, Mirabel is _____ and Bruno is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk neutral, risk averse.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Exercise 1 - Risk Aversion and Certainty Equivalent

Jackie owns shares of a company. With probability 80%, the value of the investment will go up to $x = \$200$, and with probability 20% the value of the investment will go down to $x = \$50$. Jackie's utility from income is given by the function

$$u(x) = \sqrt{x}$$

- a) Compute the expected value of the investment and Jackie's expected utility from holding this investment.
- b) Compute the certainty equivalent of this investment. In other words, the certain amount that makes Jackie indifferent between holding the investment or selling it for that certain amount (the certain amount provides the same expected utility as the investment).
- c) Compute the risk premium associated with this investment.

Question 12:

A consumer owns shares of a company. With probability 80% the value of the investment will go up to $x = \$144$, and with probability 20% the value of the investment will go down to $x = \$81$. The consumer's utility from income is given by the function

$$u(x) = \sqrt{x}.$$

- a) Compute the consumer's expected utility from holding this investment.
- b) Compute the expected value of this investment.
- c) Compute the certainty equivalent of this investment.
- d) Compute the risk premium associated with this investment.

Question 14:

Using the information from the previous question, compute the associated risk premium.

- (A) 3
- (B) 1
- (C) 7
- (D) 10
- (E) 4

Question 15:

Tom has the following utility from income: $u(I) = \sqrt{I}$. He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to

Question 21:

Arthur has the following utility from income:

$$u(I) = \sqrt{I}.$$

He has a collection of rare comic books that is worth \$10,000. With probability 50% a silverfish invasion will destroy all his books, bringing the value of his collection to zero. With probability 50% nothing occurs, and his collection keeps its full value. A company called Insect Invasion Insurance (III) offers to partially insure the collection for a price $P = \$1,900$. That is, Arthur first pays the insurance premium $P = \$1,900$; and then, in case of a silverfish invasion, III will send to Arthur a \$2,044 check. If Arthur buys this partial insurance plan, then his expected utility will be _____. Therefore, Arthur _____.

[Be careful with your computation. Keep as many decimal places as possible along all the steps of the problem. Once you are done with all computations, round your final answer to the nearest integer. It should be clear which one.]

- (A) 53 ; should buy the insurance
- (B) 51 ; should buy the insurance
- (C) 51 ; should not buy the insurance
- (D) 53 ; should not buy the insurance
- (E) None of the above, because Arthur is indifferent between buying the insurance or not.

Question 22:

Feng has all her wealth invested in Company A and in Company B. These are agricultural companies, and their value can go up or down depending on the amount of rainfall this season. There is a 50% chance that it

will rain a lot, in which case the value of her investment in Company A will go up to \$100 while her investment in Company B will go up to \$90. There is a 50% chance of a drought, in which case the value of her investment in Company A will go down to \$80, while her investment in Company B will go down to \$70.

Nan has her wealth invested in Company C and in Company D. Company C is an airline company, while Company D is an oil company. Their value can go up or down depending on the international price of oil. There is a 50% chance that the price of oil will go up, in which case her investment in Company D will go up to \$100 while her investment in Company C will go down to \$70. There is a 50% chance that the price of oil will go down, in which case the value of her investment in Company D will go down to \$80, while her investment in Company C will go up to \$90.

- a) Are the values of Company A and Company B positively or negatively correlated?
- b) Are the values of Company C and Company D positively or negatively correlated?
- c) Who has the riskier investment portfolio, Feng or Nan?

Question 23:

Mark wants to buy a new car to become a rideshare driver. He can buy a regular car or an electric car. His profit depends on the type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$120	\$90

There is a 50/50 chance that the gas price will be low or high. Mark wants to maximize his expected profit.

- a) Should Mark purchase a Regular Car or an Electric Car?

Question 24:

Mark wants to buy a new car to become a rideshare driver. He can buy a regular car or an electric car. His profit depends on the type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$102	\$90

There is a 50/50 chance that the gas price will be low or high. Mark wants to maximize his expected profit.

- a) Should Mark purchase a Regular Car or an Electric Car?

b)